LogLens

SIEM-Lite Security Monitoring — Minor Project Report

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1. PROJECT OVERVIEW

LogLens is a SIEM-lite web application that analyzes Apache/Nginx access logs, detects common web attacks, and visualizes security findings through an interactive dashboard.

2. OBJECTIVES

- Parse web-server access logs accurately.
- Detect SQL Injection, XSS, Directory Traversal, and Brute Force attacks.
- Display comprehensive security metrics.
- Export report findings in CSV and PDF formats.

3. TECHNOLOGY STACK

Python | Flask | React | Vite | Recharts | Regular Expressions | ReportLab

4. WORKFLOW

Upload Log → Parse → Threat Detection → Dashboard → CSV/PDF Export

5. THREAT DETECTION PATTERNS

Attack Type	Detection Rule / Pattern
SQL Injection	' OR 1=1, UNION SELECT
Cross-Site Scripting (XSS)	<script>, javascript:
Directory Traversal	../ patterns
Brute Force	Repeated HTTP 401 responses from a single IP address

6. KEY FEATURES & RESULTS

Key Features: Interactive dashboard, attack timeline, threat summary table, analytical charts, and CSV/PDF report generation.

Results: The application successfully detected supported attacks from sample logs and generated visual reports for security analysis.

7. FUTURE SCOPE & CONCLUSION

Future Scope: Expand support for additional log formats, real-time monitoring, Geo-IP enrichment, threat intelligence integration, and ML-assisted anomaly detection.

Conclusion: LogLens demonstrates a practical, SIEM-lite approach for automated web log analysis using signature-based and behavioral detection mechanisms.